THESIS SUPERVISOR'S REPORT

I. IDENTIFICATION DATA

Thesis title:	**Self-Supervised Segmentation of Synaptic Structures in Fluorescence Microscopy**
Author's name:	**Veronika Anokhina**
Type of thesis :	bachelor
Faculty/Institute:	Faculty of Electrical Engineering (FEE)
Department:	Department of Computer Science, FEE CTU Prague
Thesis reviewer:	Ing. Vojtěch Brejtr
Reviewer's department:	CIIRC - Cognitive systems and neurosciences

II. EVALUATION OF INDIVIDUAL CRITERIA

Assignment — **extraordinarily challenging**

How demanding was the assigned project?

The assignment required the student to build an end-to-end self-supervised segmentation and clustering pipeline for synaptic puncta in confocal fluorescence microscopy, without any manually annotated data. The task demanded integrating multiple non-trivial technical components: self-supervised pretraining (SimMIM, VICReg) of a SwinUNETR encoder, pseudo-label generation from classical image analysis, supervised fine-tuning on those pseudo-labels, unsupervised clustering with Leiden, and statistical validation with PERMANOVA. Beyond the ML pipeline, the assignment called for a production-ready Python package, Dockerized deployment, and a Django web interface. For a bachelor's thesis this scope is demanding, especially given that the dataset (802 real confocal acquisitions) was live research data collected in parallel with the thesis work, not a pre-packaged benchmark.

Fulfilment of assignment — **fulfilled**

How well does the thesis fulfil the assigned task? Have the primary goals been achieved? Which assigned tasks have been incompletely covered, and which parts of the thesis are overextended? Justify your answer.

All primary goals were met. The pipeline operates fully without manual annotations, the encoder pretraining comparison across three initializations was carried out systematically, pseudo-labels from both LoG and Spotiflow were implemented and contrasted, and the clustering stage yielded statistically significant treatment-group separation (PERMANOVA $F = 14.9$, $R^2 = 0.42$, $p = 0.005$ for the MoBY encoder). The Python package, Docker image, and Django web interface are all present and functional. The thesis honestly flags the one open gap: segmentation quality cannot be quantified without a hand-annotated subset that does not yet exist. This is not a shortcoming of execution but a real limitation of the current dataset, and the student addresses it explicitly in the Discussion and Future Work chapters.

Activity and independence when creating final thesis — **A - excellent.**

Assess whether the student had a positive approach, whether the time limits were met, whether the conception was regularly consulted and whether the student was well prepared for the consultations. Assess the student's ability to work independently.

The student showed strong initiative from the outset. She was involved in the confocal data acquisition sessions, developed all preprocessing scripts, and made all major design choices independently after consultation. Consultations were well-prepared and productive. She managed a $\sim$1 TB raw dataset on distributed infrastructure (MetaCentrum/e-INFRA CZ) without difficulty. The pace of delivery was appropriate throughout.

Technical level — **A - excellent.**

Is the thesis technically sound? How well did the student employ expertise in his/her field of study? Does the student explain clearly what he/she has done?

The thesis demonstrates solid command of the methodological stack. The student correctly adapts SwinUNETR from its original 3D volumetric setting to 2D fluorescence patches and justifies the design choices, 2D MIP over 3D processing, foreground-restricted reconstruction loss to avoid background-mean collapse, fluorescence-safe augmentation policy excluding gamma and brightness jitter, pseudo-FDT preprocessing to regularize the patchy dendrite channel before Frangi filtering. The joint SimMIM + VICReg objective is well motivated by the observation that reconstruction loss alone collapses encoder rank; the trade-off between "val_recon_fg" and "RankMe" is analyzed correctly. Using PERMANOVA on cluster-frequency vectors as an extrinsic downstream metric, rather than relying solely on the pseudo-label Dice, shows good methodological judgment. The segmentation evaluation is honestly scoped: the student recognizes that Dice against pseudo-labels cannot distinguish a good detector from one that has memorized the LoG output and says so explicitly. Furthermore, the student developed a fully functional Python package with a web interface.

Formal level and language level, scope of thesis **A - excellent.**

Are formalisms and notations used properly? Is the thesis organized in a logical way? Is the thesis sufficiently extensive? Is the thesis well-presented? Is the language clear and understandable? Is the English satisfactory?

The thesis is written in fluent, clear English with only occasional minor stylistic lapses. The structure is logical (introduction → theory → experiments → software → discussion). Figures are well designed and informative. The bibliography (76 entries) is comprehensive and correctly formatted. The main formal weaknesses are: (1) some figures are referenced in the text before they appear in the document (e.g., Figure 3.3 is discussed in Section 3.4 but rendered after the clustering section); (2) a handful of equation labels are inconsistent; (3) the Discussion chapter, while analytically sound, is somewhat brief relative to the experimental content. None of these lower the overall readability by a significant amount.

Selection of sources, citation correctness **A - excellent.**

Does the thesis make adequate reference to earlier work on the topic? Was the selection of sources adequate? Is the student's original work clearly distinguished from earlier work in the field? Do the bibliographic citations meet the standards?

The literature coverage is appropriate and up to date, including the key references for all machine learning, pharmacological and microscopy-based parts of the paper. The student clearly distinguishes her own contributions from prior work throughout. Citation style is consistent and meets standard academic requirements.

Additional commentary and evaluation (optional)

Comment on the overall quality of the thesis, its novelty and its impact on the field, its strengths and weaknesses, the utility of the solution that is presented, the theoretical/formal level, the student's skillfulness, etc.

The overall level of the bachelor thesis is significantly higher than what would be expected of a bachelor student. The completed package has a high likelihood of being used in future research projects undertaken by the team at NÚDZ.

III. OVERALL EVALUATION, QUESTIONS FOR THE PRESENTATION AND DEFENSE OF THE THESIS, SUGGESTED GRADE

Summarize your opinion on the thesis and explain your final grading.

The thesis delivers a complete, reproducible, and scientifically relevant pipeline for annotation-free synaptic puncta segmentation and pharmacological treatment-group analysis, applied to a large real-world confocal dataset from an ongoing neuroplasticity study at NÚDZ. The student demonstrated genuine mastery of the full stack, from domain-specific signal processing through state-of-the-art self-supervised learning to production software engineering. The results (PERMANOVA $R^2 = 0.42$, $p = 0.005$) are meaningful for the research group and the software deliverables are immediately usable.

The grade that I award for the thesis is **A - excellent.**

Date: **26.5.2026**

Signature: